

Newman-Penrose Formalism Calculations

Stephani et al., Equation (16.19)

Metric

$$\begin{aligned}
 ds^2 = & \left(-\frac{\omega_p \cosh(c) \sin((R - R_m)\omega_m) - \omega_m \sin((R - R_p)\omega_p) \sinh(c)}{\omega_p \cosh(c) \sin((R - R_m)\omega_m) + \omega_m \sin((R - R_p)\omega_p) \sinh(c)} \right) dt \otimes dt + \left(\frac{\omega_p \cosh(c) \sin((R - R_m)\omega_m) + \omega_m \sin((R - R_p)\omega_p) \sinh(c)}{\omega_p \cosh(c) \sin((R - R_m)\omega_m) - \omega_m \sin((R - R_p)\omega_p) \sinh(c)} \right) dR \\
 & \otimes dR + \left(\frac{\omega_p^2 \cosh(c)^2 \sin((R - R_m)\omega_m)^2 + 2\omega_m \omega_p \cosh(c) \sin((R - R_m)\omega_m) \sin((R - R_p)\omega_p) \sinh(c) + \omega_m^2 \sin((R - R_p)\omega_p)^2 \sinh(c)^2}{\omega_m^2 \omega_p^2} \right) d\theta \otimes d\theta \\
 & + \frac{\left(\omega_p^2 \cosh(c)^2 \sin((R - R_m)\omega_m)^2 + 2\omega_m \omega_p \cosh(c) \sin((R - R_m)\omega_m) \sin((R - R_p)\omega_p) \sinh(c) + \omega_m^2 \sin((R - R_p)\omega_p)^2 \sinh(c)^2 \right) \sin(\theta)^2}{\omega_m^2 \omega_p^2} d\phi \otimes d\phi
 \end{aligned}$$

Coordinates

$$(x^\mu) = (t, R, \theta, \phi)$$

Metric Functions

$$A = \frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m}$$

$$B = \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p}$$

$$Z = \frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)} - \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)}$$

$$V = \frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} - \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p}$$

$$W = \frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p}$$

Null Tetrad

Covariant components

$$l = \left(-\sqrt{\frac{1}{2}} \sqrt{\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)} - \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)}} \right) dt$$

$$+ \left(\frac{\sqrt{2}}{2 \sqrt{\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)} - \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)}}} \right) dR$$

$$n = \left(-\sqrt{\frac{1}{2}} \sqrt{\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)} - \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)}} \right) dt$$

$$+ \left(-\frac{\sqrt{2}}{2 \sqrt{\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)} - \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p \left(\frac{\cosh(c) \sin(R\omega_m - R_m\omega_m)}{\omega_m} + \frac{\sin(R\omega_p - R_p\omega_p) \sinh(c)}{\omega_p} \right)}}} \right) dR$$

$$m = \left(\frac{\sqrt{2} \cosh(c) \sin(R\omega_m - R_m\omega_m)}{2\omega_m} + \frac{\sqrt{2} \sin(R\omega_p - R_p\omega_p) \sinh(c)}{2\omega_p} \right) d\theta + \left(\frac{i\sqrt{2} \cosh(c) \sin(R\omega_m - R_m\omega_m) \sin(\theta)}{2\omega_m} + \frac{i\sqrt{2} \sin(R\omega_p - R_p\omega_p) \sin(\theta) \sinh(c)}{2\omega_p} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2} \cosh(c) \sin(R\omega_m - R_m\omega_m)}{2\omega_m} + \frac{\sqrt{2} \sin(R\omega_p - R_p\omega_p) \sinh(c)}{2\omega_p} \right) d\theta + \left(-\frac{i\sqrt{2} \cosh(c) \sin(R\omega_m - R_m\omega_m) \sin(\theta)}{2\omega_m} - \frac{i\sqrt{2} \sin(R\omega_p - R_p\omega_p) \sin(\theta) \sinh(c)}{2\omega_p} \right) d\phi$$

Contravariant components

$$l = \left(\frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m)} + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \right) dt$$

$$+ \left(\frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m)} + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \right) dR$$

$$n = \left(\frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m)} + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \right) dt$$

$$+ \left(- \frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m)} + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \right) dR$$

$$m = \left(\frac{\omega_m \omega_p}{\sqrt{2} \omega_p \cosh(c) \sin(R\omega_m - R_m \omega_m) + \sqrt{2} \omega_m \sin(R\omega_p - R_p \omega_p) \sinh(c)} \right) d\theta + \left(\frac{i \omega_m \omega_p}{(\sqrt{2} \omega_p \cosh(c) \sin(R\omega_m - R_m \omega_m) + \sqrt{2} \omega_m \sin(R\omega_p - R_p \omega_p) \sinh(c)) \sin(\theta)} \right) d\phi$$

$$\overline{m} = \left(\frac{\omega_m \omega_p}{\sqrt{2} \omega_p \cosh(c) \sin(R\omega_m - R_m \omega_m) + \sqrt{2} \omega_m \sin(R\omega_p - R_p \omega_p) \sinh(c)} \right) d\theta + \left(-\frac{i \omega_m \omega_p}{(\sqrt{2} \omega_p \cosh(c) \sin(R\omega_m - R_m \omega_m) + \sqrt{2} \omega_m \sin(R\omega_p - R_p \omega_p) \sinh(c)) \sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = \frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m)} + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c) \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \partial_t$$

$$+ \frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m)} + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c) \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \partial_R$$

$$\Delta = \frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \partial_t$$

$$- \frac{\sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}}{\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)} \partial_R$$

$$\delta = \frac{\omega_m \omega_p}{\sqrt{2} \omega_p \cosh(c) \sin(R \omega_m - R_m \omega_m) + \sqrt{2} \omega_m \sin(R \omega_p - R_p \omega_p) \sinh(c)} \partial_\theta + \frac{i \omega_m \omega_p}{(\sqrt{2} \omega_p \cosh(c) \sin(R \omega_m - R_m \omega_m) + \sqrt{2} \omega_m \sin(R \omega_p - R_p \omega_p) \sinh(c)) \sin(\theta)} \partial_\phi$$

$$\alpha = -\frac{\omega_m \omega_p \cos(\theta)}{2(\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)) \sin(\theta)}$$

$$\beta = \frac{\omega_m \omega_p \cos(\theta)}{2(\sqrt{2}\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \sqrt{2}\omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)) \sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\omega_m \omega_p^2 \cos(R\omega_p - R_p\omega_p) \cosh(c) \sin(R\omega_m - R_m\omega_m) \sinh(c)}{4(\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c))^{\frac{3}{2}} \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}} + \frac{\sqrt{2}\omega_m^2 \omega_p \cos(R\omega_m - R_m\omega_m) \cosh(c) \sin(R\omega_p - R_p\omega_p) \sinh(c)}{4(\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c))^{\frac{3}{2}} \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}}$$

$$\gamma = -\frac{\sqrt{2}\omega_m \omega_p^2 \cos(R\omega_p - R_p\omega_p) \cosh(c) \sin(R\omega_m - R_m\omega_m) \sinh(c)}{4(\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c))^{\frac{3}{2}} \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}} + \frac{\sqrt{2}\omega_m^2 \omega_p \cos(R\omega_m - R_m\omega_m) \cosh(c) \sin(R\omega_p - R_p\omega_p) \sinh(c)}{4(\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) + \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c))^{\frac{3}{2}} \sqrt{\omega_p \cosh(c) \sin(R\omega_m - R_m\omega_m) - \omega_m \sin(R\omega_p - R_p\omega_p) \sinh(c)}}$$

Ricci Tensor Components

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$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"